Dated:

$$S = \sqrt{\frac{\Sigma(x-\bar{x})^2}{n-1}}$$

# Formula Sheet

## ★ Estimation

E

• Z: $\bar{x} - Z_{\alpha/2}\frac{s}{\sqrt{n}} < \mu < \bar{x} + Z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$  $n = \left(\frac{Z_{\alpha/2} \times \sigma}{E}\right)^2$

• t: $\bar{x} - t_{\alpha/2, df}\frac{s}{\sqrt{n}} < \mu < \bar{x} + t_{\alpha/2, df}\frac{s}{\sqrt{n}}$ ; $df = n-1$

• Difference between Two Means using z: $(\bar{x}_1 - \bar{x}_2) \pm Z_{\alpha/2}\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$

•

• " " " " " t: $(\bar{x}_1 - \bar{x}_2) \pm t_{\alpha/2, df}\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$

- where $df = \min(n_1 - 1, n_2 - 1)$ (Unequal Variance)

• " " // // using t (Equal Variance):

- pooled Variance = $S_p^2 = \frac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{n_1+n_2-2}$ ; $df = n_1 + n_2 - 2$
- $(\bar{x}_1 - \bar{x}_2) \pm t_{\alpha/2, df}\sqrt{\frac{S_p}{n_1} + \frac{S_p}{n_2}}$

## ★ Hypothesis Testing

• $Z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$  • If p-value < $\alpha$, reject, otherwise accept $H_0$

• Two Mean $Z = \frac{(\bar{x}_1 - \bar{x}_2) - D_0}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$

## ★ Gradient Descent

$\hat{y} = b_0 + b_1 x$  $\partial L/\partial b_0 = \frac{1}{n}\sum_{n}^{i}(b_0 + b_1 x - y)$

$L = MSE = \frac{1}{2n}\Sigma(b_0 + b_1 x - y)^2$  $\partial L/\partial b_1 = \frac{1}{n}\sum(b_0 + b_1 x - y)x$

$b_0 new = b_0 old - \alpha \cdot (\partial L/\partial b_0)$

$b_1 new = b_1 old - \alpha \cdot (\partial L/\partial b_1)$

## ★ Linear Regression

• $\hat{y} = b_0 + b_1 x$  • $S_{xx} = \Sigma x_i^2 - \frac{(\Sigma x_i)^2}{n}$  • $s^2 = \frac{SSE}{n-2}$  $(\Sigma x, y, x^2, xy)$

• $b_1 = \frac{S_{xy}}{S_{xx}}$  • $S_{xy} = \Sigma x_i y_i - \frac{(\Sigma x_i)(\Sigma y_i)}{n}$

• $S_{yy} = \Sigma y_i^2 - \frac{(\Sigma y_i)^2}{n}$  • $e = y - \hat{y}$ (Residuals)

• $b_0 = \bar{y} - b_1\bar{x}$  • $r = \frac{S_{xy}}{\sqrt{S_{xx} S_{yy}}}$

• $t = \frac{b_1 - \beta_{00}}{s\sqrt{\frac{\Sigma x_i^2}{n S_{xx}}}}$  • $t = \frac{b_1 - \beta_{10}}{s/\sqrt{S_{xx}}}$  • $t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$

• $SSE = S_{yy} - b_1 S_{xy}$

Slope $\beta_1$ ↳ • $b_1 \pm t_{\alpha/2}\left(\frac{s}{\sqrt{S_{xx}}}\right)$  Intercept $\beta_0$ ↳ • $b_1 \pm t_{\alpha/2}\, s\sqrt{\frac{\Sigma x_i^2}{n S_{xx}}}$

Dated:

* ANOVA
  - $SST = \Sigma x_i^2 - (\Sigma x_i)^2/n$ $\qquad v_1 = k-1$
  - $SSA = \Sigma(T_j^2/n_j) - (\Sigma x_i)^2/n$ $\qquad v_2 = k(n-1)$ OR $N-k$
  - $SSE = SST - SSA$

$$F = \frac{MS(SSA)}{MS(SSE)}$$

* Hypothesis
  * Paired t-test
    - $d_i = x_{1i} - x_{2i}$ $\qquad \bullet\ t = \dfrac{\bar{d} - d_0}{s_d/\sqrt{n}}$ $\qquad \bar{d} \pm t_{\alpha/2, df} \dfrac{s_d}{\sqrt{n}}$
    - $\bar{d} = \dfrac{\Sigma d_i}{n}$
    - $s_d = \sqrt{\dfrac{\Sigma(d_i - \bar{d})^2}{n-1}}$ $\qquad \bullet\ df = n-1$

*Dated:*

***Discrete**

- $E(X) = \sum_x x P(x)$
- $Var(X) = E(X-\mu)^2$
  $= \sum_x (x-\mu)^2 P(x)$
  $= \sum_x x^2 P(x) - \mu^2$

- $Cov(X,Y) = E(X-\mu_x)(Y-\mu_y)$
  $= \sum_x \sum_y (x-\mu_x)(y-\mu_y) P(x,y)$
  $= \sum_x \sum_y (xy) P(x,y) - \mu_x \mu_y$

***Continuous**

- $E(X) = \int x f(x)\,dx$
- $Var(X) = E(X-\mu)^2$
  $= \int (x-\mu)^2 f(x)\,dx$
  $= \int x^2 f(x)\,dx - \mu^2$

- $Cov(X,Y) = E(X-\mu_x)(Y-\mu_y)$
  $= \iint (x-\mu_x)(y-\mu_y) f(x,y)\,dx\,dy$
  $= \iint (xy) f(x,y)\,dx\,dy - \mu_x \mu_y$

- $Z$ = Standard Normal Random Variable
- $\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$, Std Norm pdf
- $\Phi(x) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi}} e^{-z^2/2}\,dz$, std Norm cdf

Continous:

PDF. $\int_{-\infty}^{\infty} f(x)\,dx = 1$

CDF $\int_{-\infty}^{x} f_x(t)\,dt$

- Shortcut for E(X) & Var(X)

$E(X) = n \cdot \frac{K}{N}$ $\quad$ $Var(X) = n \cdot \frac{K}{N} \cdot \frac{N-K}{N} \cdot \frac{N-n}{N-1}$

- Hypergeometric Formula

$P(X=k) = \frac{(\text{ways to pick errors}) \times (\text{ways to pick non-errors})}{(\text{ways to pick } n \text{ blocks})}$

- Joint Probability Mass function:

$P = \frac{n!}{x_1!\,x_2!\,x_3!} * p_1^{x_1} \times p_2^{x_2} \times p_3^{x_3}$

- Curve Z-Areas $<$: directly
  $>$: 1-

$Z = \frac{X-\mu}{\sigma}$

Cont.

*** Uniform Dist**

- PDF: $\begin{cases} \frac{1}{B-A} & A \le x \le B \\ 0, & \text{otherwise} \end{cases}$
- CDF $\begin{cases} 0, & x < A \\ \frac{x-A}{B-A}, & A \le x \le B \\ 1, & x > B \end{cases}$

**Dated:**

# Formulas

* Binomial

$\mu = np$

$\sigma^2 = npq$

* Poisson

$P(X=x) = \dfrac{(e^{-\mu} \cdot \mu^x)}{x!}$

$\mu = \sigma^2 = \lambda t$

rate $\lambda$ over time $t$

* Normal

$Z = \dfrac{X-\mu}{\sigma}$

* Uniform

$f(x) = \dfrac{1}{b-a}$

$\mu = \dfrac{a+b}{2}$

$\sigma^2 = \dfrac{(b-a)^2}{12}$

$P(c \le x \le d) = \dfrac{d-c}{b-a}$

- $E(X) = \mu = \sum x \cdot p(x)$
- $Var(X) = E[X^2] - E(X)^2$

- Bayes Theorem

$$P(B_i \mid A) = \frac{P(A \mid B_i) \cdot P(B_i)}{\sum P(A \mid B_j) \cdot P(B_j)}$$

* Continuous

- PMF $= \int_{-\infty}^{\infty} f(x)\,dx = 1$ $\quad \begin{cases} \frac{1}{B-A}, & A \le x \le B \\ 0, & \text{else} \end{cases}$

- CDF $= \int_{-\infty}^{x} f(t)\,dt$ $\quad \begin{cases} 0, & \cancel{\text{else}}\ x < A \\ \frac{x-A}{B-A}, & A \le x \le B \\ 0, & x > B \end{cases}$

- Independance

- $P(X,Y) == P(X) \cdot P(Y)$

if so, then

$E(XY) = E(X) \cdot E(Y)$

$Var(X+Y) = Var(X) + Var(Y)$

* $Var(aX+b) = a^2 \cdot Var(X)$

- Covariance $(X,Y) = E[XY] - E[X] \cdot E[Y]$
- Corelation $= \rho(X,Y) = \dfrac{Cov(X,Y)}{\sigma_x \cdot \sigma_y}$

* Hypergeometric formula

$$P(X=k) = \frac{(\text{ways to pick err}) \times (\text{ways to pick errors})}{(\text{ways to pick } n \text{ blocks})}$$

* Mean & Variance (Continuous)

- $E(X) = \mu = \int_{-\infty}^{\infty} x \cdot f(x)\,dx$
- $Var(X) = E(x^2) - E(x)^2$ where $\int x^2 f(x)\,dx$

Dated:

# Formula Sheet

## * Expectation and Variance (5)

- $E(X) = \sum x \cdot P_x$
- $Var(X) = E(X^2) - [E(X)]^2$

  where $E(X^2) = \sum x^2 \cdot P(x)$

## * Key Prop.

- $E(aX+b) = aE(X) + b$
- $Var(aX+b) = a^2 Var(X)$
- $Var(X_1 + X_2) = Var(X_1) + Var(X_2)$

## • PMF, CDF, Joint, Marginal, Independance (4)

- PMF = $P(X=x)$, $\sum P(X=x) = 1$, "single"
- CDF = $F(x) = P(X \leq x)$, "at most"
- Joint = $P(X=x, Y=y)$, when two var occur tgt
- Marginal = $P(X=x) = \sum_y P(X=x, Y=y)$, "dist of x only" "Ignore other var"
- Independance = $P(X,Y) = P(X)P(Y)$
- Total Prob law = $P(A) = \sum P(A|B_i)P(B_i)$
- Conditional Prob = $P(A|B) = P(A \cap B)/P(B)$
- Bayes' Theorem = $P(B_i|A) = \dfrac{P(A|B_i)P(B_i)}{\sum P(A|B_j)P(B_j)}$

(6)

## * Poisson Distribution

- $\lambda$ = Avg No. of events
- $P(x) = \dfrac{e^{-\lambda}\lambda^x}{x!}, x = 0, 1, 2, \ldots$
- $E(X) = \lambda$
- $Var(X) = \lambda$
- $P(x; \lambda t) = \dfrac{e^{-\lambda t}(\lambda t^x)}{x!}, x = 0, 1, 2, \ldots$

(6)

## * Binomial Distribution

$n$ = number of trials

$p$ = probability of success

$b(x; n, p) = P(X) = \binom{n}{x} p^x q^{n-x}$

$E(X) = np$

$Var(X) = npq$

## * Uniform, Normal Dist (e.g. $f(x) = 2x^{-3}$ for $x \geq 1$)

- $\mu = E(X) = \int x f(x)\,dx = \int_1^{\infty} 2x^{-2}\,dx = \left| -2x^{-1} \right|_1^{\infty} = -0 - (-2) = 2$

(7)

## * Uniform Distribution

- $(a, b)$ = ranges of values
- $f(x) = \dfrac{1}{b-a}, a < x < b$
- $E(X) = \dfrac{a+b}{2}$
- $Var(X) = \dfrac{(b-a)^2}{12}$

(7)

## * Normal Distribution

- $\mu$ = expectation, location parameter
- $\sigma$ = std, scale parameter
- $f(x) = \dfrac{1}{\sigma\sqrt{2\pi}} \exp\left\{\dfrac{-(x-\mu)^2}{2\sigma^2}\right\}, -\infty < x < \infty$
- $E(X) = \mu$
- $Var(X) = \sigma^2$
- $Z = \dfrac{X-\mu}{\sigma}$

Note: ($\mu = 0$ & $\sigma = 1$ is called std Norm dist)